Module 5: Enzymes

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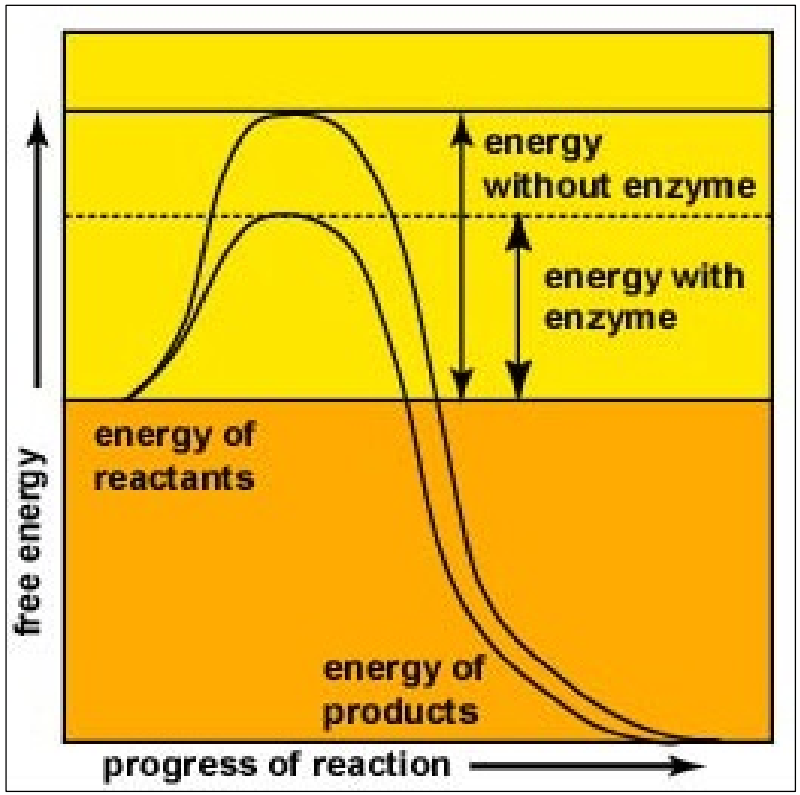
Introduction

- **Definition:** Beadle (1948) defined enzymes as indispensable compounds that play a key role in metabolism by bringing direction and control to the physiological process of living cells.
- **Enzymology** is a branch of biochemistry which deals with the study of enzymes, their kinetics, structure, and function, as well as their relation to each other.
- Enzymes are biological catalysts i.e., **biocatalysts**, which help in lowering activation energy of a chemical reaction inside our body. Any change in enzyme complement of living cell is immediately reflected in the change in the physiological and biochemical process of the cell.



Characteristics of Enzymes

- Enzymes are generally globular proteins.
- Exception: Some RNA molecules called **ribozymes** can also be enzymes. These are usually found in the nuclear region of cells and catalyze the splitting of RNA molecules.
- Many enzymes require **cofactors** for their catalytic activity. The cofactor may be a complex organic molecule called a **coenzyme** or a metal ion such as Fe^{2+}, Mn^{2+}, Zn^{2+}, or Mg^{2+}.
- When a coenzyme or metal ion is tightly bound through the covalent bond with the enzyme protein, it is called a **prosthetic group**.
- **Isozymes** (also known as isoenzymes) are enzymes that differ in amino acid sequence but catalyze the same chemical reaction. Example: lactate dehydrogenase (LDH), which is used to catalyze the synthesis of glucose in anaerobic metabolism of glucose.

Co-enzymes and co-factors

Coenzyme	Precursor vitamin	Catalytic role
Biocytin	Biotin (Vit B7)	Transfer of CO2
Flavin adenine dinucleotide (FAD)	Riboflavin (Vit B2)	Transfer of electrons
Coenzyme A	Pantothenic acid (Vit B3)	Transfer of acyl & alkyl group
Nicotinamide adenine dinucleotide (NAD)	Niacin (Vit B5)	Transfer of hydride

Metal ions	Enzyme name
Fe2+ or Fe3+	Catalase, peroxidase, cytochrome oxidase
Cu2+	Cytochrome oxidase
Mg2+	DNA polymerase
Mn2+	Arginase
K+	Pyruvate kinase
Mo2+	Nitrogenase, nitrate reductase
Zn2+	Carbonic anhydrase, alcohol dehydrogenase
Ni2+	Urease

Classification

According to the Nomenclature Commission of the International Union of Biochemistry, enzymes are arranged into **six groups** according to the general type of reaction they catalyze.

1. Oxidoreductases
2. Transferases
3. Hydrolases
4. Lyases
5. Isomerases
6. Ligases

$$AH_2 + B \rightleftharpoons A + BH_2$$

Oxidation-reduction reaction

$$AX + B \rightleftharpoons A + BX$$

Transfer of groups

$$A - B + H_2O \rightleftharpoons A - H + BOH$$

Hydrolytic reaction

$$\begin{array}{ccc} X & Y & \\ | & | & \\ A = B & \rightleftharpoons & A = B \end{array} \quad \begin{array}{cc} Y & X \\ | & | \\ & \end{array}$$

Transfer of group within molecule to yield isomeric form

$$A + B \rightleftharpoons A - B$$

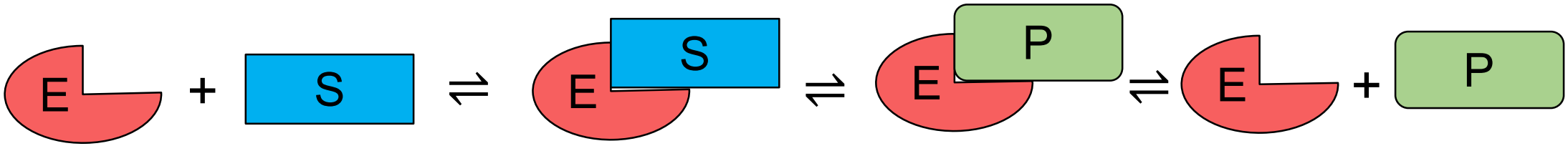
Condensation of two molecule coupled through ATP hydrolysis

$$\begin{array}{c} X \quad Y \\ | \quad\ \ | \\ A = B + X - Y \rightleftharpoons A - B \end{array}$$

Addition or removal of double bonds

Theories of Enzyme Action

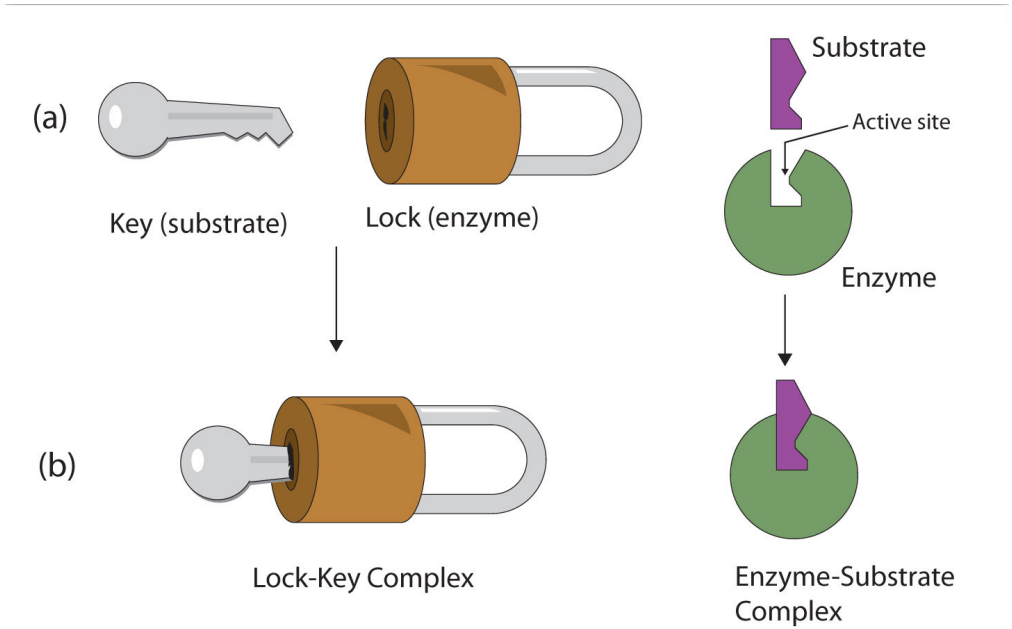
- The enzymes act on a substrate by combining with the substrate molecules to form an enzyme-substrate complex.
- Enzymes (E) have specific active sites for the attachments of substrate (S) molecule where an enzyme can form intimate relationship with the substrate.
- The binding of enzyme and substrate is highly specific. A given enzyme usually binds to only one kind of substrate.



- The enzyme-substrate combination (ES) brings about a deformation in some of the bonds in substrate molecule, which leads to the formation enzyme-product (EP) complex.
- Subsequently, the product (P) is released from the enzyme.

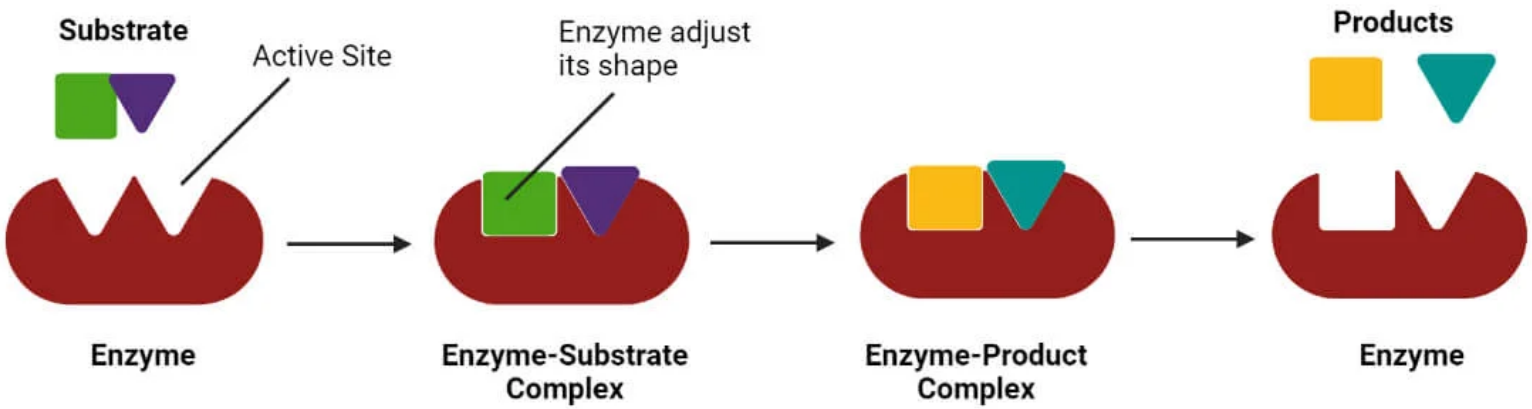
Lock and Key Theory

- It was proposed by **Fisher (1894)**.
- According to this theory, the enzyme and the substrate both have **strictly complementary structures** which during complex formation fit to each other like a specific key in a particular lock.
- Only a specific key (substrate) with correctly positioned teeth fits into the keyhole (active site) of the lock (enzyme).
- This theory gave a satisfactory explanation for enzyme action, however, this theory **fails to explain the flexible nature of an enzyme.**



Induced Fit Theory

- Proposed by **Koshland (1966).**
- According to this model, the enzyme and substrate do not have strictly complementary structures, but the enzyme has **a flexible active site** with changes according to substrate configuration.
- This model can be compared with hand and glove. the glove adjusts itself in the hands of those who wear it. Similarly, the substrate induces **conformational changes** in the enzyme during enzyme action.



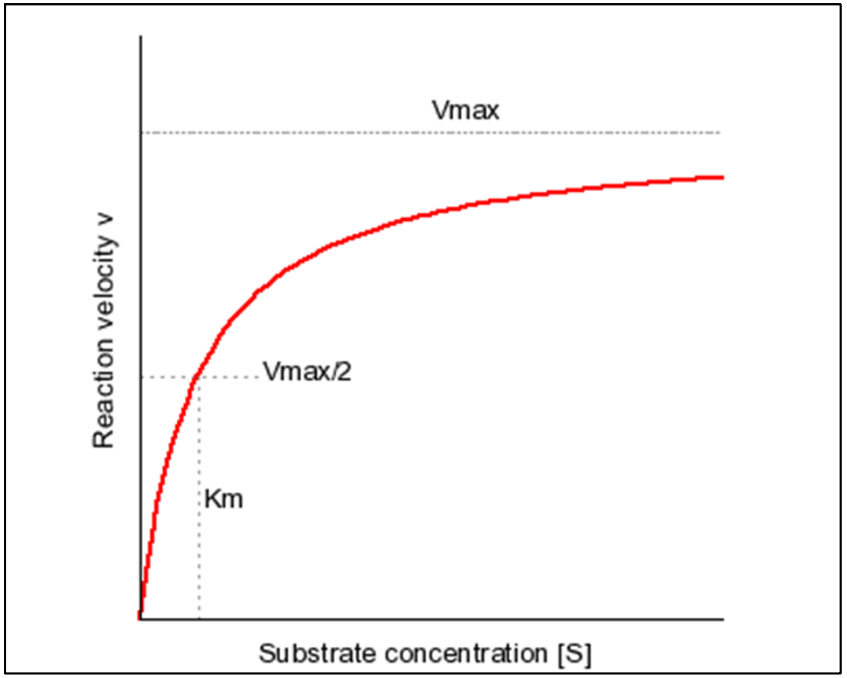
Kinetics of Enzyme Action

- **Leonor Michaelis** and **Maud Menten** in 1913 explained the kinetics of enzyme-catalyzed reactions.
- According to their theory, ES complex is an intermediate during catalysis.
- Michaelis and Menten expressed it as a single order equation, thus it is called **Michaelis-Menten equation**.

$$V_0 = \frac{V_{max}\,[S]}{K_m + [S]}$$

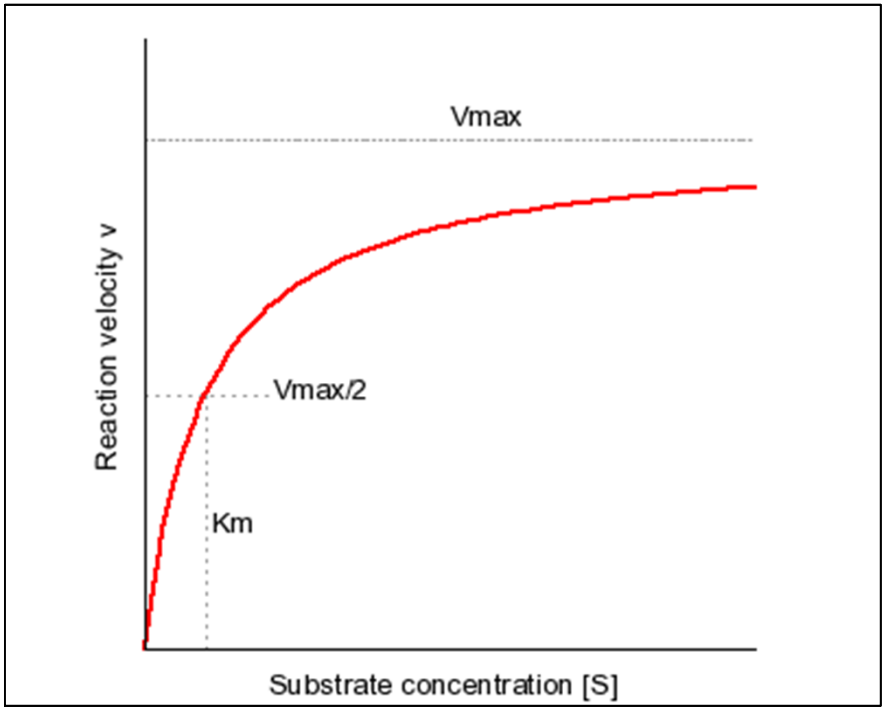
Michaelis constant

V_0 = Initial velocity of reaction
V_{max} = Maximum velocity of reaction
[S] = Substrate concentration
K_m = Substrate concentration when $V_{max}/2$



Kinetics of Enzyme Action

- From the graph, when [S]<< K_m, $V_0=(V_{max}/K_m)/[S]$, i.e., the reaction rate is **directly proportional** to substrate concentration.
- When [S]>>K_m, $V_0 = V_{max}$, i.e., the reaction rate is **maximum** and independent of substrate concentration.
- When [S]= K_m, then V0= V_{max} /2.
- The maximum velocity, V_{max}, represents the turnover number of an enzyme. **Turnover number** is the number of substrate molecules converted into the product by an enzyme molecule in a unit of time when the enzyme is fully saturated with substrate.
- Turnover number = K_{cat}

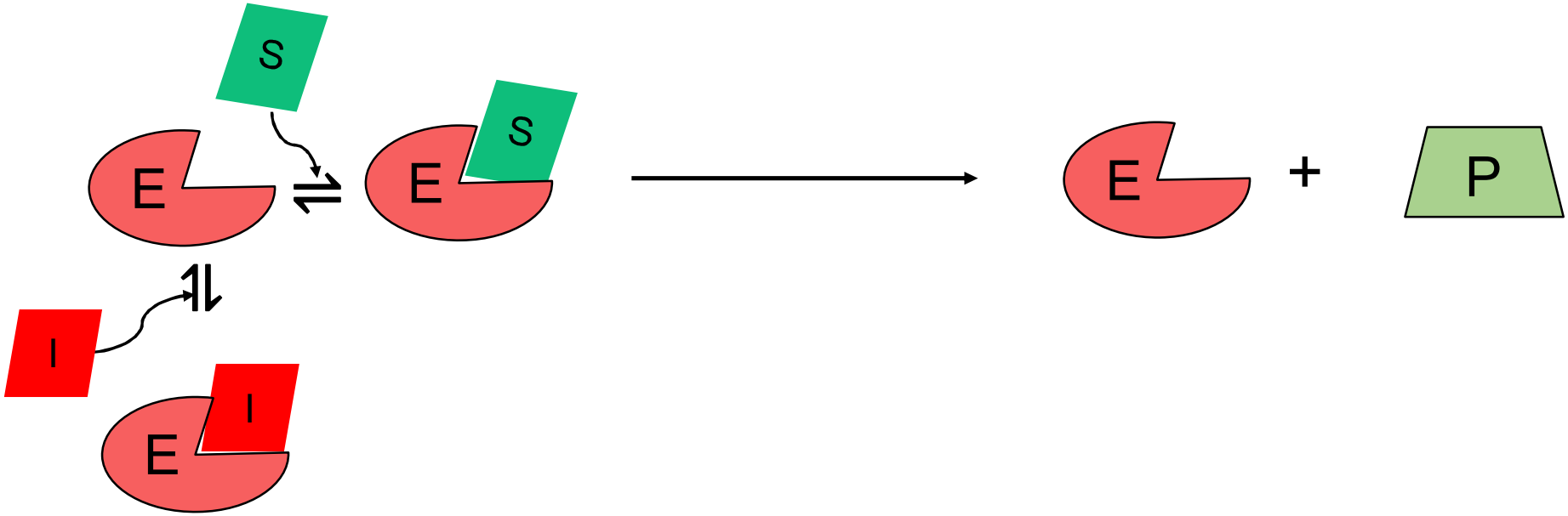


Inhibition of Enzyme Action

- Substances that decrease the rate of an enzyme-catalyzed reaction are called **enzyme inhibitors**, and the process is known as **enzyme inhibition**.
- Enzyme inhibition can be classified as **irreversible inhibition** and **reversible inhibition**.

1. **Irreversible inhibition:** The inhibitor binds tightly to the enzyme and does not dissociate. Example: The antibiotic **penicillin** acts as an inhibitor and binds with the enzyme **transpeptidase**, which synthesizes the bacterial cell wall. Hence, binding this drug to the enzyme **prevents cell wall synthesis**, thus killing the bacteria.
2. **Reversible inhibition:** The inhibitor rapidly dissociates from the enzyme-inhibitor complex. There are **three** types of reversible inhibitions: **competitive**, **non-competitive**, and **uncompetitive**.

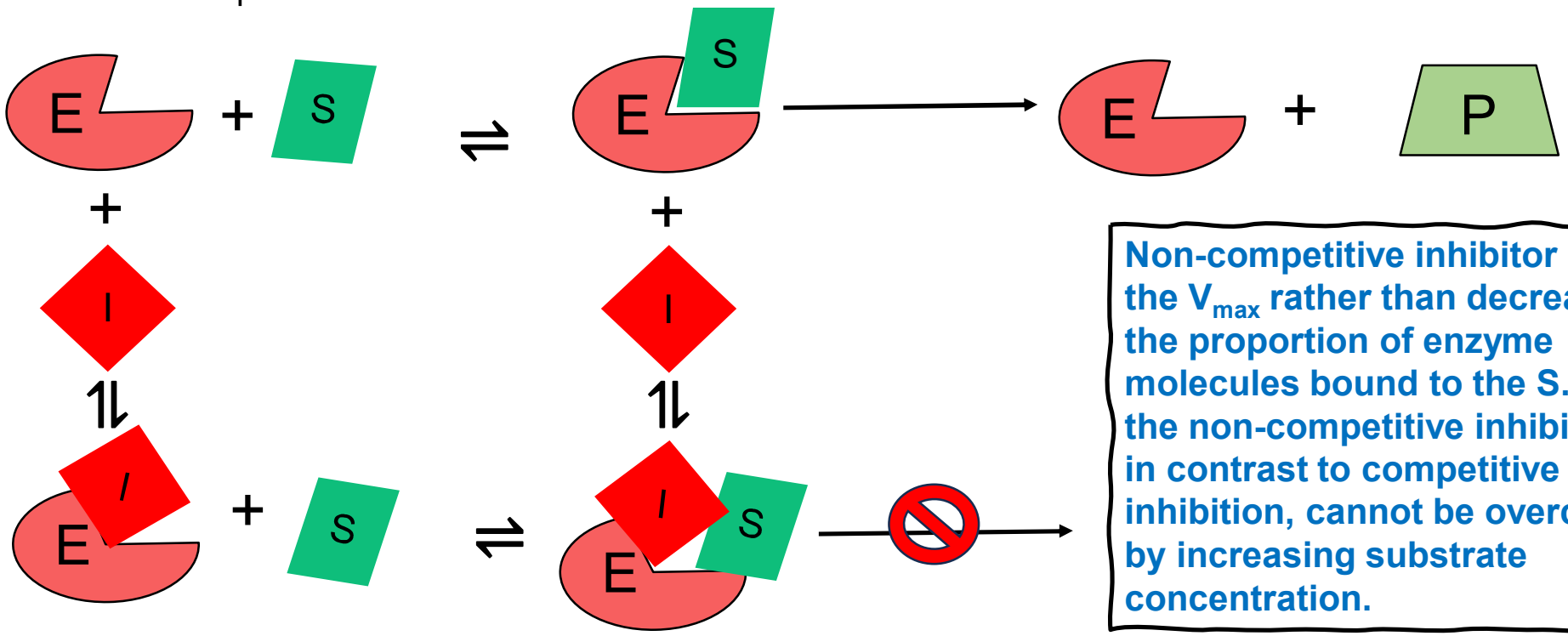
Competitive Inhibition



- Because the inhibitor (I) structurally resembles the substrate (S), the enzyme can form enzyme-substrate ES complex or it can form enzyme-inhibitor EI complex but not both ESI.
- Competitive inhibitors decrease the rate of reaction by reducing the amount of active enzyme molecules bound to a substrate.

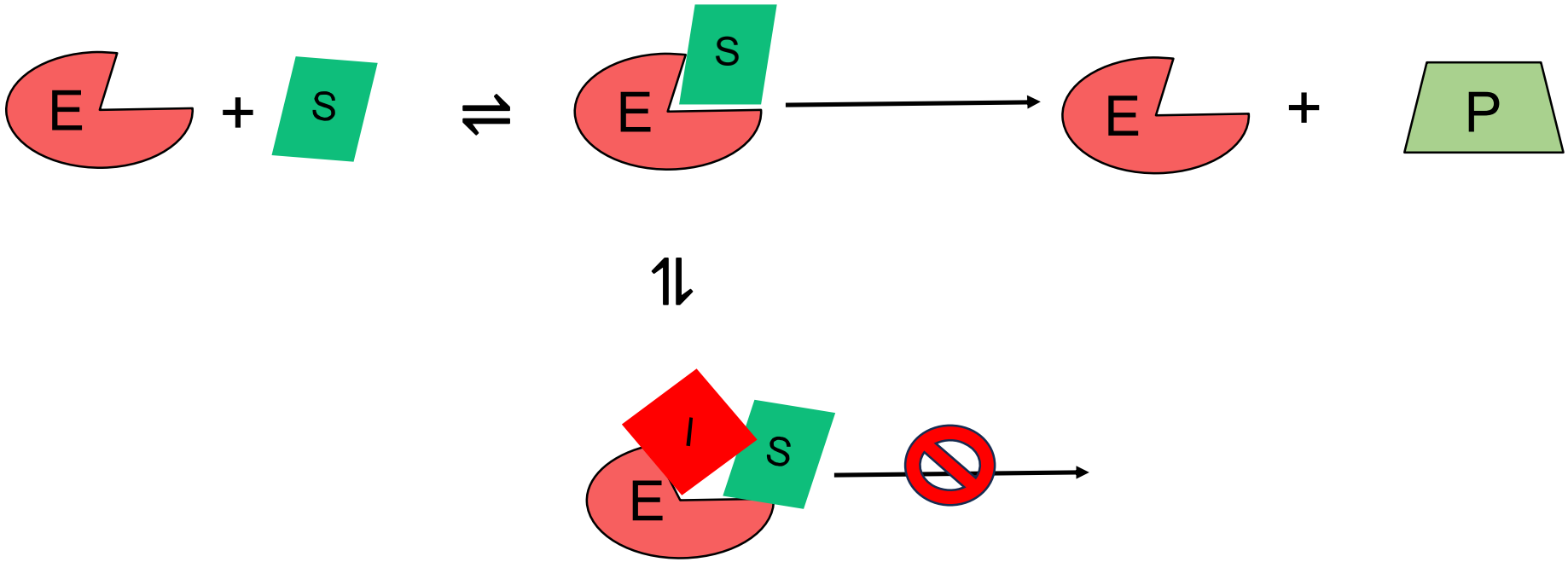
Non-competitive Inhibition

- Inhibitor has no structural similarity with substrate and binds with the enzyme at different site other than the active site.
- No competition between S and I, and formation of ES, EI and ESI takes place.
- The inhibitor I and substrate S can bind simultaneously to the same enzyme molecule as their binding sites are different and hence do not overlap.



Non-competitive inhibitor lowers the V_{max} rather than decreasing the proportion of enzyme molecules bound to the S. Thus, the non-competitive inhibition, in contrast to competitive inhibition, cannot be overcome by increasing substrate concentration.

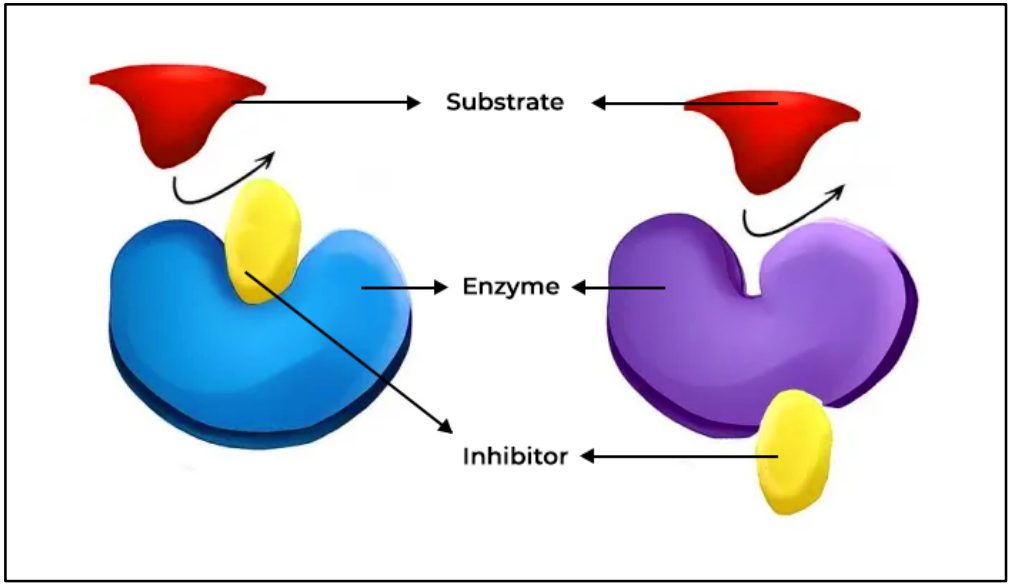
Uncompetitive Inhibition



- The inhibitor does not bind to the free enzyme.
- It binds only to the enzyme-substrate (ES) complex directly, or its binding is facilitated by the conformational change after the substrate binds to an enzyme.

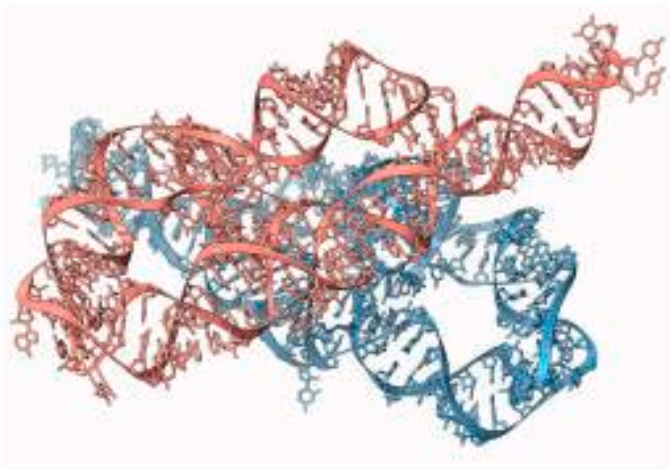
Allosteric Enzymes

- Enzymes that have **an additional binding site** for **effector molecules** other than the active site. The binding brings about conformational changes, thereby changing its catalytic properties.
- Allosteric enzymes **do not** obey Michaelis-Menten kinetics.
- Many allosteric enzymes **take part in the biochemical pathways** so that the system is well controlled and modulated such as Aspartate Transcarbamoylase (ATCase), **Glucokinase**, and Acetyl-CoA Carboxylase



Ribozymes

- Ribozymes (**ribonucleic acid enzymes**) are RNA molecules capable of catalyzing specific biochemical reactions, similar to the action of protein enzymes.
- It contributed to the **RNA world hypothesis**, which suggests that RNA may have been essential in the evolution of prebiotic self-replicating systems.
- The most common activities of natural or in vitro-evolved ribozymes are the **cleavage or ligation of RNA and DNA and peptide bond formation**.
- Within the ribosome, ribozymes function as part of the large subunit ribosomal RNA to link amino acids during protein synthesis. They also participate in various RNA processing reactions, including RNA splicing, viral replication, and transfer RNA biosynthesis.
- Example: The hammerhead ribozyme, the VS ribozyme, Leadzyme, and the hairpin ribozyme.



https://www.youtube.com/watch?v=yk14dOOvwMk

Thank you.